UTP Universiti Teknologi PETRONAS — energising futures

GRAND FINALE · HEALTHTECH TRACK

CODEFEST'26 -SYNTECH-

Kawan.

kawan (Malay) · *friend*

An AI guardian for people who live alone.

It notices, checks in, and alerts family only when needed.

Not a doctor. · Not a surveillance camera. · **A guardian.**

Team Nexora UTP CODEFEST '26

Ajmal Haikal bin Mohd Zakhiruddin · Nik Mohamad Fairuz Addailami bin Mohamed Zabedi
Muhammad Iyad Mustaqim bin Muhd Gaddafi · Muhammad Zulfiqar bin Zuraimi



01 The problem

IMPACT

15.6% of Malaysian adults have **diabetes**, 29.2% hypertension; ~4 in 10 don't know it.[1]

684k people die from **falls** each year; most fatal falls are in over-60s.[2]

80% of fallers over 90 **couldn't get up alone**; 30% lay on the floor ≥ 1 hour.[3]

~50% of long-term patients **don't take medicine** as prescribed.[4]

02 Our solution

IMPACT · USABILITY

01 Notice
Camera spots falls and unusual stillness; tracks medicines and readings.

02 Check in
"**Are you okay?**" by voice, button or chat, in English or Malay.

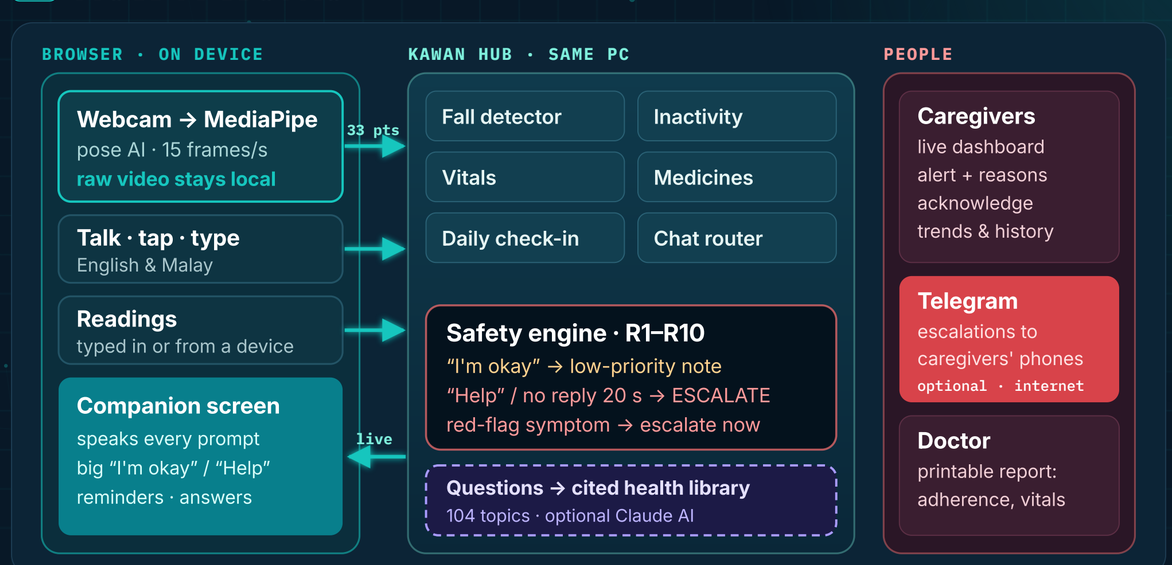
03 Escalate only if needed
"Help", a red-flag symptom or **no reply in 20 s** alerts caregivers.

04 Explain why
Caregivers see **why**, and what else it could be.

Conventional alarm (PENDANT / PANIC BUTTON)	Person falls → They must press the button → ✕ Hurt or confused: alarm mostly unused[3]
Kawan (NO PANIC BUTTON · NO WEARABLE)	Webcam notices the fall → "Are you okay?" → ✓ "Help" or no reply in 20 s → caregivers alerted

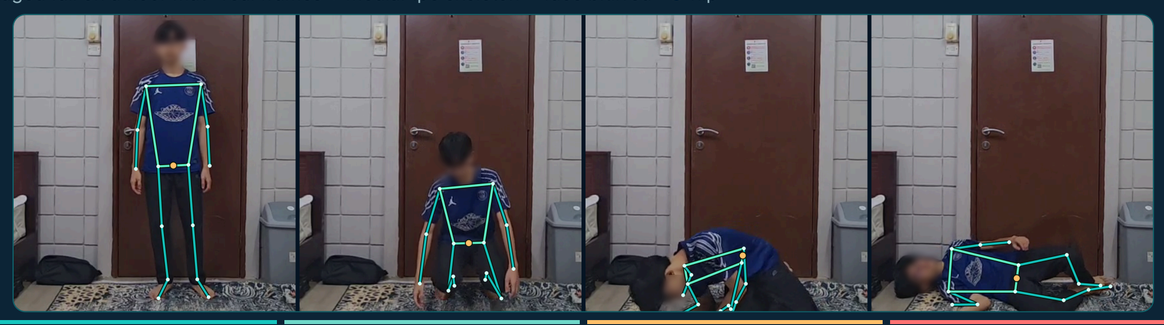
03 How it works

TECHNICAL EXECUTION



Our own fall test, seen by Kawan

Staged fall on a floor mat · real frames + MediaPipe skeleton[6] · face blurred · ● hip



1 · Upright 1.0 s

2 · Hips drop ≥ 0.6 torso in ≤ 1.5 s

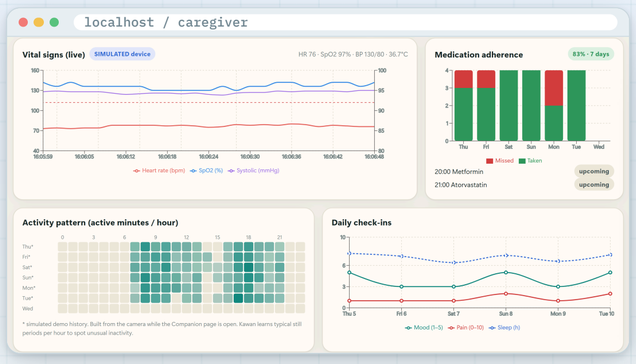
3 · Horizontal torso ≥ 60°

4 · Still down 2 s later → check in

`Kawan's real output: possible fall at 4.7 s. Hip dropped 18% of frame (0.7× torso) in 1.4 s · torso 68° · no recovery within 2 s`

04 The app

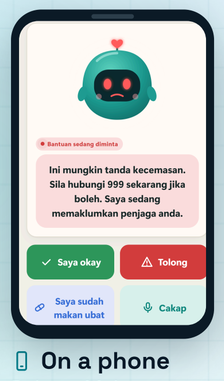
USABILITY · REAL SCREENS, DEMO DATA



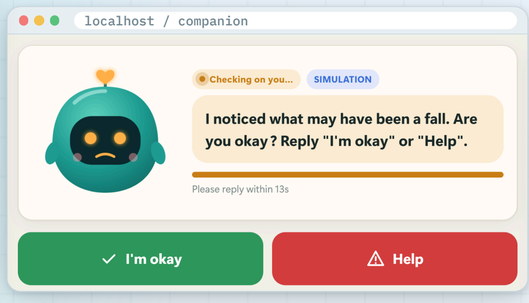
Caregiver dashboard
Vitals, medicine adherence, activity pattern, daily check-ins.



Fall alert, with reasons
Why it fired, what else it could be.



On a phone
Bahasa Melayu



Companion check-in
Big buttons, spoken prompt, 20 s timer.

ALSO BUILT

- Spoken reminders
- Sustained-reading vitals
- Daily check-in triage
- 104 cited topics, EN & BM
- Doctor report

05 Safe & private by design

- **Rules decide, AI only explains.** Check-in replies never go to the AI.
- **Fails safe.** "Okay, but my leg hurts" counts as help.
- **Explains itself:** every alert says why, and what else it could be.

WHAT LEAVES THE BROWSER

Camera video **stays in the browser** → **33 landmarks** x, y, visibility · 15/s → Kawan hub **same PC by default**

No video or images are sent or stored. **One opt-in exception:** "Check my medicine" sends one photo to Claude when tapped (needs an AI key).

06 Offline & online

Works without internet
- Pose detection, fall & inactivity rules
- Check-ins by button and chat
- Spoken prompts (device voices)
- Reminders, vitals, daily check-in
- 104-topic health library
- Dashboard & report (on the hub PC)

Needs internet
- Voice input (browser speech service)
- Telegram alerts to phones
- Claude AI answers & pill check (optional, API key)
- Phones & remote caregivers (secure link)

If the internet drops: detection and check-ins keep running on the hub PC; Claude answers fall back to the offline library. Phone alerts and voice input stop until it returns.

07 Results

TECHNICAL EXECUTION · USABILITY

PRELIMINARY TEST · FALL DETECTOR ON RECORDED CLIPS[5]

90% — **27 / 30** **fall clips detected.** The earlier frame-size rule: 17/30.

12 / 40 **everyday clips triggered a check-in**; 10 show people lying down on purpose.

A check-in asks the person → "I'm okay" = low-priority note → Phones alerted only on "help" or no reply

Test conditions: UR Fall dataset · 70 lab clips of actors · one room, side camera · same model and settings as the app · last frame held 10 s (person stays down).

⚠ **Not real-world accuracy.** Preliminary lab testing; not clinically validated.

148 TESTS

Automated tests

117 Backend · Python (pytest)
19 Frontend · TypeScript (Vitest)
12 End-to-end · real browser (Playwright)

GitHub Actions runs all 148 on every push.

EARLY INFORMAL FEEDBACK · 4 TESTERS (3 FRIENDS AND A PARENT)

> *"The reminders are the part I'd use every day."*

> *"I don't need to read anything. I just talk to it and it talks back."*

> *"It noticed when I fell and asked if I was okay. That's reassuring when no one's home."*

> *"I asked what ibuprofen is and how it works, and finally understood it."*

WE HEARD: **Alerts on 2–3 caregivers' phones**
✓ **Done:** Telegram to every caregiver

WE HEARD: **Voice sometimes misses words**
Tap buttons as backup; better speech next

WE HEARD: **Knows less than chatbots**
Deliberate: cited sources only

08 Feasibility

- **Hardware:** a laptop or tablet with a webcam + Chrome/Edge.
- **Cost:** free open-source stack; AI and Telegram optional.
- **Setup:** one command; a secure link for phones.

Roadmap & next steps

✓ NOW
Working prototype; tried by 4 early users.

NEXT
Caregiver login · hosted multi-household version · push alerts · device integration.

LATER
Clinical validation with healthcare partners · more emergency types.

REFERENCES

1. IPH, MOH Malaysia. *NHMS 2023: NCD Fact Sheet.* 2024.
2. WHO. *Falls: Fact Sheet.* 2021.
3. Fleming J, Brayne C. *Inability to get up after falling.* BMJ 2008;337:a2227.
4. WHO. *Adherence to Long-Term Therapies.* 2003.
5. Kwolek B, Kepski M. *UR Fall Detection Dataset.* Comput Methods Programs Biomed 2014;117:489–501.
6. Google. *MediaPipe Pose Landmarker.* Google AI Edge.

Prototype — not a medical device. In an emergency call 999. One camera, one person in view. Health library: MOH Malaysia CPGs, WHO, NHS. Built with help from Claude Code (AI coding assistant), disclosed as the rules require.

DEMO VIDEO · SOURCE CODE